

Viva- 20 [These marks will be taken as 20% of the obtained mark in the viva voce course of this semester]

2) SEE-Semester End Examination (70 marks)

Bloom's Category	Lab Final (70)
Remember	5
Understand	15
Apply	30
Analyze	20
Evaluate	
Create	

a. Course Code: ICE- 3203 Course Title: Embedded Systems and Internet of Things Course Type: Core Course Pre-requisite (if any): ICE 1103, ICE 1105, ICE 1201, ICE 2103, ICE 2107		Year/Semester: Year 3, Semester 2 Credit Value: 3 Contact Hours: 42 Total Marks: 100
Rationale	This course covers the principles, design, and applications of embedded systems and the Internet of Things (IoT). It includes an in-depth study of embedded system hardware, software, and communication protocols, with a focus on IoT applications. Students will gain practical experience through hands-on projects involving sensor integration, data acquisition, and IoT communication protocols.	
Course Objectives	• To understand the fundamental principles and architecture of embedded systems and IoT. • To develop proficiency in programming microcontrollers and interfacing various sensors and actuators. • To learn and implement standard communication protocols used in IoT systems. • To design and build IoT applications integrating embedded systems with cloud services. • To analyze and address security and privacy challenges in IoT deployments.	
b. Course Learning Outcomes (CLOs): Upon completion of this course the students will be able to-		
CLO 1	Demonstrate proficiency in designing and programming embedded systems using microcontrollers.	
CLO 2	Integrate and interface various sensors and actuators with embedded systems.	
CLO 3	Implement and utilize standard communication protocols for IoT applications.	
CLO 4	Develop IoT applications that connect embedded systems to cloud platforms.	
CLO 5	Analyze and address security and privacy issues in IoT systems.	

c. Mapping Course Learning Outcomes (CLOs) with the PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO 1	√		√		√		√					
CLO 2	√	√	√		√		√		√			
CLO 3	√		√		√		√					
CLO 4	√	√	√	√	√		√					
CLO 5	√	√	√	√	√	√	√	√				√

d. Lesson Plan (as per week) or Mapping Course Learning Outcomes (CLOs) with course content, Teaching-Learning and Assessment strategy:

Sl. No.	Summary of course contents	Hrs.	Alignment to CLOs
1	Introduction to Embedded Systems and IoT Topics: - Definition and Characteristics of Embedded Systems - Overview of IoT: Definition, Architecture, and Applications - Differences between Embedded Systems and IoT Readings: "Embedded Systems: Real-Time Interfacing to Arm Cortex-M Microcontrollers" by Jonathan W. Valvano, Chapter 1 "Internet of Things: A Hands-On Approach" by Arshdeep Bahga and Vijay Madisetti, Chapter 1	4	CLO 1, CLO 2
2	Embedded System Hardware Topics: - Microcontrollers: Architecture, Selection Criteria - Sensors and Actuators: Types and Interfacing Techniques - Memory: RAM, ROM, Flash, and EEPROM Readings: "The 8051 Microcontroller and Embedded Systems" by Muhammad Ali Mazidi and Janice Gillispie Mazidi, Chapter 3 "Microcontroller Theory and Applications with the PIC18F" by M. Rafiquzzaman, Chapter 4 Assignment: - Design a basic embedded system using a microcontroller and various sensors.	5	CLO 1, CLO 2
3	Embedded System Software Topics: - Embedded C Programming: Basics, GPIO, Timers, and Interrupts - Real-Time Operating Systems (RTOS): Concepts and Applications - Debugging and Testing Embedded Software Readings: "Embedded C Programming and the Microchip PIC" by Richard Barnett, Larry O'Cull, and Sarah Cox, Chapter 5 "MicroC/OS-II: The Real-Time Kernel" by Jean 4J. Labrosse, Chapter 2 Assignment: - Write and debug a C program to control an LED using a microcontroller.	5	CLO 1, CLO 2
4	Communication Protocols for Embedded Systems Topics: - Serial Communication: UART, SPI, I2C - Wireless Communication: Bluetooth, Wi-Fi, Zigbee - Networking Basics: TCP/IP, MQTT, CoAP Readings: "Serial Port Complete" by Jan Axelson, Chapter 4 "Wireless Communications and Networks" by William Stallings, Chapter 8 Assignment: - Implement a simple communication protocol between two microcontrollers.	4	CLO 3
5	IoT Architecture and Protocols Topics: IoT System Architecture: Sensors, Edge Devices, Cloud	4	CLO 3, CLO 4

	- IoT Communication Protocols: MQTT, CoAP, HTTP/HTTPS - IoT Standards and Frameworks: IEEE, IETF, OneM2M Readings: "Building the Internet of Things" by Maciej Kranz, Chapter 3 "Designing the Internet of Things" by Adrian McEwen and Hakim Cassimally, Chapter 5 Assignment: - Develop an IoT application using MQTT to send sensor data to a cloud server.		
6	Sensor Networks and Data Acquisition Topics: - Types of Sensors: Temperature, Humidity, Motion, etc. - Data Acquisition Systems: ADC, Signal Conditioning - Wireless Sensor Networks (WSN): Architecture, Protocols Readings: "Introduction to Wireless Sensor Networks" by Anna Förster, Chapter 6 "Sensor Networks: A Tutorial" by Chee-Yee Chong and S. P. Kumar (IEEE Paper) Assignment: - Design a wireless sensor network to monitor environmental conditions.	4	CLO 2
7	IoT Data Management and Analytics Topics: - Data Storage: SQL, NoSQL Databases - Data Processing: Stream Processing, Batch Processing - Data Analytics: Machine Learning, Big Data Tools Readings: "Data Science for IoT Engineers" by Vijay Madiseti, Chapter 4 "Big Data: Principles and Best Practices" by Arun C. Murthy and Vignesh Prajapati, Chapter 7 Assignment: - Analyze sensor data using a cloud-based analytics platform.	4	CLO 4
8	Topics: - Security Challenges in IoT: Threats, Vulnerabilities - Encryption and Authentication: SSL/TLS, Public Key Infrastructure (PKI) - Privacy Concerns and Solutions: Data Anonymization, Access Control Readings: "Security and Privacy in Internet of Things (IoTs): Models, Algorithms, and Implementations" by Fei Hu, Chapter 3 "IoT Security: Advances in Authentication" by Parikshit N. Mahalle, Chapter 2 Assignment: - Implement a secure communication channel for an IoT device using encryption.	4	CLO 5
9	IoT Platforms and Tools Topics: - IoT Development Platforms: Arduino, Raspberry Pi, ESP8266 - Cloud Platforms for IoT: AWS IoT, Google Cloud IoT, Azure IoT - Development Tools: IDEs, Debuggers, Simulators Readings: "Getting Started with the Internet of Things" by Cuno Pfister, Chapter 2 "Programming the Internet of Things" by Andy King, Chapter 4	4	CLO 4, CLO 5

	Assignment: Develop an IoT prototype using a development platform and cloud service.		
10	Emerging Trends in Embedded Systems and IoT Topics: • Edge Computing: Concepts and Applications • AI and Machine Learning in IoT: Use Cases and Tools • Future Trends: 5G, Blockchain, Digital Twins Readings: • "Edge Computing: A Primer" by K. R. Jackson, Chapter 2 • "Artificial Intelligence for IoT Cookbook" by Michael Roshak, Chapter 5 Assignment: • Research and present a case study on an emerging trend in IoT.	4	CLO 1, CLO 3, CLO 4

e. Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO 1	Lecture	Quiz and Class Test
CLO 2	Lecture and Group discussion	Viva voce and Final Exam
CLO 3	Problem based learning and Presentation	Assignment and Final Exam
CLO 4	Lecture	Choral responses and Reflective writing
CLO 5	Problem based learning	Assignment and Viva voce

f. Reference Books:

1. "Embedded Systems: Real-Time Interfacing to Arm Cortex-M Microcontrollers" by Jonathan W. Valvano
2. "Internet of Things: A Hands-On Approach" by Arshdeep Bahga and Vijay Madisetti
3. "The 8051 Microcontroller and Embedded Systems" by Muhammad Ali Mazidi and Janice Gillispie Mazidi
4. "Microcontroller Theory and Applications with the PIC18F" by M. Rafiquzzaman
5. "Building the Internet of Things" by Maciej Kranz

g. Assessment Pattern

1) Continuous Assessment (30 Marks)

Bloom's Category	Class Tests (20)	Assignments (5)	Class Participation (5)
Remember	5		5
Understand	5		
Apply	5		
Analyze	5		
Evaluate		2	
Create		3	

2) Summative (70 Marks)

Bloom's Category	Semester Final (70)
Remember	20
Understand	25
Apply	25
Analyze	20
Evaluate	5
Create	5

a. Course Code: ICE 3204 Course Title: Embedded Systems and Internet of Things Sessional Course Type: Core Pre-requisite (if any): ICE 3203		Year/Semester: Year 3, Semester 2 Credit Value: 1.5 Contact Hours: 42 Total Marks: 100
Rationale	This course covers the principles, design, and applications of embedded systems and the Internet of Things (IoT). It includes an in-depth study of embedded system hardware, software, and communication protocols, with a focus on IoT applications. Students will gain practical experience through hands-on projects involving sensor integration, data acquisition, and IoT communication protocols.	
Course Objectives	● To understand the fundamental principles and architecture of embedded systems and IoT. ● To develop proficiency in programming microcontrollers and interfacing various sensors and actuators. ● To learn and implement standard communication protocols used in IoT systems. ● To design and build IoT applications integrating embedded systems with cloud services. ● To analyze and address security and privacy challenges in IoT deployments.	
b. Course Learning Outcomes (CLOs): Upon completion of this course the students will be able to-		
CLO 1	Demonstrate proficiency in designing and programming embedded systems using microcontrollers.	
CLO 2	Integrate and interface various sensors and actuators with embedded systems.	
CLO 3	Implement and utilize standard communication protocols for IoT applications.	

c. Mapping Course Learning Outcomes (CLOs) with the PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO 1	√	√										
CLO 2			√									
CLO 3				√								
CLO 4			√									
CLO 5					√							

d. Lesson Plan (as per week) or Mapping Course Learning Outcomes (CLOs) with course content, Teaching-Learning and Assessment strategy:

Sl. No.	Summary of course contents	Hrs.	Alignment to CLOs
1	Introduction to Embedded Systems and IoT: Characteristics, Definitions, Architectures, Applications	3	CLO 1
2	Embedded System Hardware: Microcontrollers, Sensors and Actuators, Memory Types	5	CLO 1, CLO 2
3	Embedded System Software: Embedded C, RTOS, GPIO, Timers, Interrupts	5	CLO 1, CLO 2
4	Communication Protocols: UART, SPI, I2C, Bluetooth, Wi-Fi, Zigbee	5	CLO 2, CLO 3
5	IoT Architecture & Protocols: MQTT, CoAP, HTTP, Edge/Cloud Integration	5	CLO 3, CLO 4
6	Sensor Networks & Data Acquisition: ADC, Signal Conditioning, WSN	4	CLO 2, CLO 3
7	IoT Data Management & Analytics: SQL/NoSQL, ML, Big Data Tools	4	CLO 3, CLO 4
8	IoT Security and Privacy: Threats, Encryption, PKI, Access Control	4	CLO 5

9	IoT Platforms and Tools: Arduino, Raspberry Pi, ESP, AWS IoT, GCP, IDEs	4	CLO 1, CLO 4
10	Emerging Trends: Edge Computing, AI in IoT, Blockchain, Digital Twins	3	CLO 4, CLO 5
11	Final Lab Exam (Lab Tasks and Viva)		

e. Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO 1	Lecture	Quiz, Viva voce and Lab Test
CLO 2	Lecture and Group discussion	Quiz, Viva voce and Lab Test
CLO 3	Lecture, Problem based learning	Quiz, Viva voce and Lab Test
CLO 4	Project-based Learning, Cloud Integration Tasks	Lab Test, Project Evaluation, Viva Voce
CLO 5	Case Study Analysis, Group Discussion	Presentation, Report, Viva Voce

f. Reference Books:

- Jonathan W. Valvano, Embedded Systems: Real-Time Interfacing to Arm Cortex-M Microcontrollers
- Arshdeep Bahga and Vijay Madisetti, Internet of Things: A Hands-On Approach
- Muhammad Ali Mazidi and Janice Gillispie Mazidi, The 8051 Microcontroller and Embedded Systems
- Fei Hu, Security and Privacy in Internet of Things (IoTs): Models, Algorithms, and Implementations
- Vijay Madisetti, Data Science for IoT Engineers.

g. Assessment Pattern	
1) Continuous Assessment (30 Marks)	
	Attendance- 10
	Viva- 20 [These marks will be taken as 20% of the obtained mark in the viva voce course of this semester]
2) SEE-Semester End Examination (70 marks)	
Bloom's Category	Lab Final (70)
Remember	5
Understand	15
Apply	30
Analyze	20
Evaluate	
Create	

a. Course Code: ICE 3205 Course Title: Computer Networks Course Type: Core Pre-requisite (if any): ICE 2105		Year/Semester: Year 3, Semester 2 Credit Value: 3 Contact Hours: 42 Total Marks: 100
Rationale	The course intends to provide an introduction to computer networks, with a special focus on internet architecture and protocols. Topics include layered network architectures, addressing, naming, forwarding, routing, communication reliability, the client-server model, web, and email protocols.	
Course Objectives	- To build an understanding of the fundamental concepts of computer networking. - To familiarize the student with the basic terminology of the computer networking area. - To introduce the student to advanced networking concepts, preparing the student for entry advanced courses in computer networking.	